

GUNA, India

WMO: 425590

Lat: 24.650N

Lon: 77.317E

Elev: 1568

StdP: 13.88

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.2	(c) 48.5	(d) 31.6	(e) 27.5	(f) 69.4	(g) 34.0	(h) 30.3	(i) 66.3	(j) 7.2	(k) 62.8	(l) 5.6	(m) 64.8	(n) 0.4	(o) 50	(p) 0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.5	109.4	73.2	107.3	72.4	105.0	71.9	81.1	90.9	80.2	89.1	79.5	88.0	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	160.1	84.9	78.3	155.6	83.8	77.6	152.0	82.8	46.6	91.1	45.5	89.1	44.7	88.1	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.6	(b) 5.4	(c) 4.9	(e) 41.0	(f) 113.1	(g) 3.1	(h) 2.0	(i) 38.8	(j) 114.6	(k) 37.0	(l) 115.7	(m) 35.2	(n) 116.8	(o) 33.0	(p) 118.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.8	62.6	68.5	78.6	87.9	94.0	90.3	82.5	80.2	81.6	79.9	72.9	65.6
	DBStd	10.30	4.53	5.32	5.47	4.45	3.54	5.91	4.09	2.62	2.91	3.36	3.62	4.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	177	105	24	0	0	0	0	0	0	0	0	1	47
	CDD50	10494	391	518	885	1137	1365	1210	1008	936	947	928	686	483
	CDD65	5196	30	123	420	687	900	760	543	471	497	463	237	65
	CDH74	65032	344	1196	4863	10101	14915	11435	5767	4102	4920	4682	2089	618
	CDH80	36073	54	389	2662	6497	10541	7396	2270	1112	1952	2333	753	114
Wind	WSAvg	1.3	1.1	1.5	1.4	1.4	1.9	2.1	1.6	1.4	1.0	0.8	0.7	0.9
Precipitation	PrecAvg	42.4	0.4	0.4	0.5	0.1	0.4	5.8	14.8	12.9	5.8	0.6	0.3	0.5
	PrecMax	77.0	1.1	2.2	2.9	0.4	1.9	31.9	25.4	26.4	24.6	2.7	3.9	6.5
	PrecMin	19.9	0.1	0.0	0.1	0.0	0.0	0.9	2.1	1.2	0.4	0.0	0.0	0.0
	PrecStd	11.3	0.3	0.6	0.7	0.1	0.5	6.5	5.6	5.2	5.3	0.8	0.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	91.5	102.6	108.6	112.1	112.3	101.6	93.7	96.9	97.7	91.3	86.2
		MCWB	62.8	64.4	67.3	69.8	74.1	73.6	77.3	78.4	73.7	70.6	66.2	64.5
	2%	DB	81.4	88.2	98.6	106.2	110.0	108.9	96.6	89.9	93.8	95.1	89.0	83.3
		MCWB	61.6	63.9	66.3	69.3	73.3	74.8	77.8	79.0	75.3	70.3	66.1	63.0
	5%	DB	78.0	85.0	95.9	104.1	108.3	105.9	93.1	87.7	91.6	93.1	86.8	80.3
		MCWB	59.8	62.7	65.4	68.7	72.6	75.0	77.9	78.5	75.8	69.9	64.9	61.4
	10%	DB	74.6	81.7	93.1	101.9	106.4	102.4	90.2	85.7	89.5	91.0	84.6	77.4
		MCWB	58.2	61.4	64.6	68.1	72.3	75.7	78.0	78.2	76.3	69.3	64.2	60.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	68.0	71.0	74.8	80.4	83.2	81.9	81.4	81.5	78.0	73.0	68.4
		MCDB	80.5	82.6	94.3	103.2	102.0	97.7	91.1	87.2	87.9	87.4	81.9	82.0
	2%	WB	63.1	65.7	68.7	71.9	78.2	81.2	80.9	80.2	79.8	76.0	70.3	64.8
		MCDB	76.7	82.2	92.6	100.5	101.9	95.1	88.7	86.1	86.7	84.9	80.4	78.8
	5%	WB	61.3	64.2	67.2	70.3	76.5	79.9	80.1	79.5	78.8	74.7	68.1	62.9
		MCDB	73.9	81.0	90.3	98.7	100.5	93.7	87.1	85.1	85.9	83.7	81.1	77.3
	10%	WB	59.7	62.6	65.8	69.0	75.0	78.8	79.3	78.8	77.9	73.0	66.3	61.2
		MCDB	71.2	79.2	88.6	97.2	99.6	92.6	85.8	84.1	84.7	83.3	80.6	75.2
Mean Daily Temperature Range	5% DB	MDBR	23.6	24.5	26.6	27.2	24.5	19.2	11.1	9.6	14.3	22.7	24.8	24.9
		MCDBR	27.4	27.8	29.3	28.9	25.9	23.0	16.9	13.6	18.7	24.8	26.5	27.5
		MCWBR	12.2	11.1	10.0	9.2	8.4	7.1	5.3	5.2	5.7	7.9	10.2	11.7
	5% WB	MCDBR	23.5	24.5	26.1	26.5	23.7	18.7	12.5	11.1	13.3	18.1	21.6	24.3
		MCWBR	11.5	11.1	10.3	9.6	9.3	6.8	5.6	5.3	5.7	7.3	9.5	11.7
Clear-Sky Solar Irradiance	taub		0.439	0.438	0.469	0.531	0.589	0.660	0.646	0.583	0.521	0.519	0.519	0.475
	taud		2.031	2.078	2.013	1.895	1.812	1.719	1.767	1.928	2.054	1.944	1.875	1.944
	Ebn at Noon		254	266	264	250	234	217	220	234	246	238	226	235
	Edh at Noon		49	49	56	64	69	75	72	61	53	56	56	51
All-Sky Solar Radiation	RadAvg		1312	1617	1977	2223	2261	1869	1327	1301	1605	1664	1395	1274
	RadStd		106	81	76	79	59	171	183	143	167	98	93	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+3.69	N/A	+0.54	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page